SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT

ALLIES CONNECT - GROUP 4

Requirement Analysis Report

Overview

This Software Requirements Specification (SRS) document defines the requirements for the Allies Connect web platform. The system is designed to connect underprivileged Georgia residents with nearby community resources, non-profit services and events while also connecting other residents to volunteer opportunities. The document establishes a shared understanding between all stakeholders by detailing the system's user types, features, operating environment, architecture, and constraints. It outlines detailed requirements for core features such as event management, volunteer management, provider portals, and administrative moderation. Additionally it addresses key characteristics such as performance, security, accessibility, and usability standards. These specifications are used to define the reference point for the design, implementation, and validation of the Phase 1 MVP deliverable for Allies Connect.

User Type Requirements

Public User

Description

A public user is any person who wishes to visit the platform in search for community resources, events and event information, and other support services. This user does not require an account.

Functional Requirements

- Search for nearby community resources
- View resources by list view or map
- Filter resources by category, date, and location
- Enter zip code or geolocation location as added resource and event filter
- View resource details

- Browse events calendar
- Filter event calendar by date and category
- Access platform via mobile or desktop device

Non-Functional Requirements

- Mobile-friendly and accessible UI
- Login is not required for browsing
- Must meet WCAG accessibility guidelines

Volunteer User

Description

A volunteer user is a public user who registers for volunteer opportunities made available by service providers

Functional Requirements

- Register a volunteer account
- Log into volunteer account
- Browse volunteer opportunities
- Enter zip code or geolocation location as added opportunity filter
- View volunteer event and role details and shift availability
- Register for volunteer event/shift
- Submit volunteer contact information
- Receive email confirmation and reminders for registered events
- Cancel registration
- Update Registration
- Access platform via mobile or desktop device

Non-Functional Requirements

- Volunteer registration must be quick and simple
- Form submission error messages must be intuitive
- Mobile-friendly and accessible UI
- Personal data must be securely stored

Service Provider

Description

A service provider is a community outreach or nonprofit organization that creates and manages resources, events, and volunteer opportunities.

Functional Requirements

- Register an organization account
- Log in to organization account
- Create and manage organization profile
- Create, manage, and remove resource listings
- Create, manage, and remove events
- Create, manage, and remove volunteer opportunities
- Define volunteer shifts and slot limits within volunteer opportunities
- View volunteer signups
- Export volunteer signup data

Non-Functional Requirements

- Provider dashboard must be mobile-friendly and easy to use
- Authentication must be secure (SSO or dual?)
- Provider data must be secure
- Provider actions must be logged
- Any data updates must be reflected on UI (defined sync time)

Admin User

Description

An admin user is a system administrator that moderates data and manages system integrity.

Functional Requirements

- Approve or reject provider registrations
- Manage listings
- Moderate content
- View system logs
- Manage user permissions
- Add additional admin users
- Add service areas
- Update service areas
- Delete service areas

Non-Functional Requirements

- Admin functions must be restricted by role
- Admin actions must be logged with time and user ID
- Admin interface must allow for easy moderation
- Admin pages must be restricted to admin users

Feature Requirements

Public Resource Finder

Description

A Google maps view that displays the local area, and features points of interest that represent resources, events, and volunteer opportunities. Users will be able to filter these events by certain metrics to view the resources they are most interested in. When a user clicks or taps on a point of interest they will be able to view more information about that opportunity.

Functional Requirements

- Uses user location to view local area
- Filter options to filter which points of interest (PoI) display on the map:
    - Name
    - Type of PoI (Resource, Event, Volunteer Opportunity, etc)
    - Services provided
        - Housing
        - Clothing
        - Food
        - Legal
        - Financial
        - Disability Support
        - Misc/Other
    - Distance
    - County
    - Date
- Display resources, events, and volunteer opportunities as points of interest
- Selecting a point of interest provides more details
- User can expand or narrow their search radius
- Events and resources are pinpointed and displayed by using the location data provided the entity's creator or manager
- User can scroll in the map view
- User can zoom in and zoom out in the map view
- PoIs are viewable as a list as well as on a map

Non-Functional Requirements

- Map Navigation should be smooth and accurate
- Filter options should be easily understood and applicable to users' needs

Priority

The public resource finder should be one of the services we build first. Having an understanding of how the different entities will be displayed on the map will give us a better idea of how to display information and lay out the options available to users.

Event Directory

Description

A calendar system for community events including free clinics, donation drives, pop-up services, and food distributions. The directory enables event creation by both administrator and providers, making it easy for residents to find accurate and up-to-date opportunities.

Functional Requirements

- Providers will have access to event creation form from their dashboard
- The provider event creation form will include the following fields:
    - Event title (`Event.title`)
    - Description (`Event.description`)
    - Category (`Event.category_id`)
    - Location (embedded geocoding) (`Event.location_id`)
    - Event Date (selection for recurring will be available) (`Event.event_date`)
    - Start date/time (start and end validation logic will be included) (`Event.start_datetime`)
    - End date/time (`Event.end_datetime`)
    - Capacity (`Event.capacity`)
    - Registration required (`Event.registration_required`)
    - Contact information (`Event.provider_id`)
    - Special instructions (`Event.special_instructions`)
- Administrators will be able to create events on behalf of organizations
- Administrators will be able to assign events to specific provider accounts
- A calendar grid will be used to display events along with a list view of the events sorted by date
- Users will be able to filter events by date range, zip code, and category
- Each event will have a detail page showing all provided information
- The event detail page will display a map showing the event location
- The event detail page will show the current status of the event that can be updated by administrators and providers who created the event
- Events will automatically be archived once the date of the event has passed
- Events will have a “Volunteers Needed” tag when appropriate

Non-Functional Requirements

- The event calendar will automatically update if the displayed information changes
- The web platform will be able to handle concurrent events without performance issues

Priority

The event directory is going to be an early target goal, potentially being developed concurrently with the public resource finder. Creating the calendar functionality will be a key resource as we continue to develop the organization panel and the volunteer management systems.

Volunteer Management System

Description

The volunteer management system allows service providers to create and manage the volunteer opportunities for their services. These take 3 main forms:

- Volunteer opportunities to assist with community events
- Volunteer opportunities that are events themselves
- Managing shifts for volunteers

Functional Requirements

- Events feature an option to request volunteer assistance
- Volunteer users can sign up for events as a volunteer
- The volunteers for an event can be seen by the event creator
- When creating an event a service provider may designate it as a "volunteer only" event
- Users that register as volunteers may view the volunteer sign up buttons, see events that are "volunteer only" and register/subscribe to a service provider for those events to automatically be added to their calendar.
- Service providers may post shifts that are available for volunteer opportunities, which they can request "subscribed" volunteers to show up for.
- The service providers should be able to make event descriptions that volunteers see with volunteer-specific information
- Administrators can create forms for the volunteers to fill out
- Organizations can export volunteer signups data to CSV files

Non-Functional Requirements

- The volunteer management system should be quick and efficient
- The volunteer management system should carefully handle the Personally Identifiable Information (PII) of Volunteers

Priority

The volunteer management system will need to be developed after the event management,and public resource systems, since many of its functions rely on tools that will be developed for those other services. It can be developed concurrently with the organization management portal, because the two will both rely on each other, so decisions will need to be made during development that affect both services.

Organization Portal

Description

The Organization Portal allows the service providers to manage their resource page, create, manage, and delete events, and organize the volunteer opportunities for their organization. This portal should be managed by a single user for an organization, or a few members that are authorized to utilize the shared login information.

Functional Requirements

- An organization registration form will be accessible to new organizations
- The organization registration form will include the following fields:
    - Organization legal name (`ServiceProvider.name`)
    - Common name (`ServiceProvider.common_name`)
    - Employer Identification Number (EIN)
    - Organization type (will be drop down selection) (`ServiceProvider.organization_type`)
    - Primary address (`ServiceProvider.address_id`)
    - Phone number (`ServiceProvider.phone_number`)
    - Website (`ServiceProvider.website`)
    - Mission statement (`ServiceProvider.mission`)
    - Service areas (drop down selection of cities and counties in GA) (`ServiceArea.label`)
    - Primary contact name (`ServiceProvider.contact_name`)
    - Primary contact email (`ServiceProvider.contact_email`)
    - Primary contact phone (`ServiceProvider.contact_phone`)
    - Password (N/A)
    - Confirm password (N/A)
- Once the form is completed, an email with a validation link will be sent to contact information provided in the organization registration form for email validation
- New organization accounts will have a PENDING status until approved by an administrator
- Providers will log in with their account email and password
- The login form will have secure password hashing
- The login form will have a password reset button that sends a link to the provided email if an account is registered with that email
- Once logged in, the portal will display a dashboard with the following information:
    - Statistics summary (resources, events, volunteers)
    - Recent activity
    - Pending approval items
    - Create resource button
    - Create event button
    - Notifications and alerts
- Providers will be able to edit organization profile information

- Organization profile information will include:
    - Logo (`ServiceProvider.logo_url`)
    - Mission statement (`ServiceProvider.mission`)
    - Website link (`ServiceProvider.website`)
    - Operating hours (`ServiceProvider.operating_hours`)
    - Languages spoken (`ServiceProvider.languages_spoken`)
    - Accessibility features (`ServiceProvider.accessibility`)
- Providers will create new resources from the organization dashboard
- The new resource form will include the following fields:
    - Title (`Resource.name`)
    - Category (`Resource.category_id`)
    - Description (`Resource.description`)
    - Address (`Resource.location_id`)
    - Operational hours (`Resource.hours`)
- Providers will be able to view list of all their organizations available resources
- Providers will be able to edit existing resources
- Providers will be able to deactivate resources
- Logs will be created with time stamps and user info whenever changes are made
- Organization dashboard will show all events, separating upcoming from past
- Providers will access volunteer management system through the organization portal

Non-Functional Requirements

- The portal will be accessible across all devices
- Industry-standard security practices will be used for authentication
- Portal will be WCAG accessible

Priority

The organization management system will be developed concurrently with the volunteer management system, since both systems share similar behaviors.

Administrator Console

Description

The administrator console allows service administrators of Allies Way to perform managerial upkeep of the Allies Way service. This includes the ability to register and approve new organizations to be added to the service, delete organizations that no longer use the service, ban users, and manage how many events organizations will be able to post to the Allies way service.

Functional Requirements

- Administrator roles will include:
    - Super Admin (Full access, approve/deny organization forms)
    - Content Moderator (modify/remove listings, resources, etc.)
- Administrators portal login will be separate from provider/volunteer logins
- Administrators console will display a queue of pending organizations
- Administrators will be able to view the full organization application
- Administrators will be able to approve/reject when viewing full organization application
- Upon approval:
    - The organization's provider account will be activated
    - Email will be sent notifying of approval with login instructions
    - Log approval action with time stamp and admin user ID
- Upon rejection:
    - Email will be sent notifying of rejection with reasoning
    - Application will be archived
    - Log rejection action with time stamp and admin user ID
- Administrators will be able to review all resources across organizations within the portal, filtering by:
    - Status
    - Organization
    - Category
    - Date
- Administrators will be able to edit any resource, event, or opportunity
- Administrators will be able to deactivate/activate content with reason
- Organization will be notified of content that is altered by an administrator
- Administration console will be able to display the following metrics:
    - Total organizations
    - Total resources by category
    - Total events
    - Total volunteer opportunities
    - Volunteer data (quantity, attended events)
- Administrators will be able to suspend organization accounts
- Administrators will be able to reset organization passwords

Non-Functional Requirements

- Administrator login must be secure to prevent data breaches
- All administrator queues and dashboards will show up to date information
- The administrator portal interface will prioritize usability and efficiency

Priority

The administrator console can be developed once the organization portal has been developed. Users with admin console access will be able to see the full spectrum of content on the app, so it only makes sense that the admin console be developed once that full context can be realized.

System Requirements

The system will be split across different server infrastructures and services. The following captures those that we will have some control over and those that are provided by third parties.

Application server. The application server will host the web server. It has the following requirements:

- Linux 9.x
- 2 CPU
- 4 GB RAM
- 40 GB Storage

Database server. The application will use a third-party database service provider. FreeSQLDatabase.com meets the requirement that all hosting be free or provided by KSU data centers.

Email server. The application will use a third-party email service provider to send email reminders about upcoming events to those who register. Gmail.com provides this service free-of-charge.

Maps functionality. The application will use a third-party service provider for mapping information. Google Maps provides this service free-of-charge.

501(c)3 validation. The application will use a third-party service to validate EIN data provided by non-profits posting to the application. GivingTuesday.org provides this service free-of-charge.

REST API. The application will be built with a REST API for access to database content and functionality.

Simple permissions model. The app will have a simple permissions model to ensure that each type of user is able to access only the data and functionality appropriate to their needs.

Constraints

- The platform must be mobile-friendly and web-based, not native mobile
- Backend must be hosted on KSU-provided infrastructure or other no-cost environments
- Accessibility should be considered (WCAG basics)
- The system must be modular enough for Phase II expansion
- No paid APIs unless provided by sponsor
- Automated features will only implemented if approved by the sponsor

Priorities

Each of the sections within the *Feature Requirements* section of this document has a *Priority* subsection. This section pulls all of these priorities together to create a list in descending order

of priority. This enables stakeholders to review the list and confirm that the order in which we plan to work on discrete elements matches their goals.

Public Resource Finder

The public resource finder is the primary reason for this web application's existence. As such, if nothing else is created, this part of the app will be.

Event Directory

The event directory is a key dependency for both the organization portal and the volunteer management system, so it needs to be implemented prior to or concurrent with those other subsystems.

Organization Portal and Volunteer Management System

The organization portal and volunteer management system have similar priority, and can be developed concurrently.

Administrator Console

The administrator console is important, but much of what it handles can be done manually in the database by a knowledgeable person, or potentially be handled by scripts that can do the work without existing as a fully-fleshed-out user interface in the web application.

Change Record

The following changes have been made since this document was originally submitted.

Feb 14, 2026

- Added a Priorities section to explain what elements are believed to have the highest priority and what elements can be dropped if time runs short.
- Added "Register a volunteer account" and "Log into volunteer account" to the Volunteer User section.
- Added "Add additional admin users" to the Admin User section.
- Added "County" and "Date" to the filter options in the Public Resource Finder section.
- Added "User can scroll in the map view," "User can zoom in and zoom out in the map view" and "PoIs are viewable as a list as well as on a map" to the Public Resource Finder section.
- Added "Events will automatically be archived once the date of the event has passed" and "Events will have a "Volunteers Needed" tag when appropriate" to the Event Directory section.

- Added “Administrators can create forms for the volunteers to fill out” and “Organizations can export volunteer signups data to CSV files” to the Volunteer Management System section.
- Added “REST API” and “Simple permission management” requirements to the System Requirements section.

Feb 15, 2026

- Removed “Filter by preferred shift slots” from the Volunteer User Functional Requirements section
- Removed “and only share what is absolutely necessary with service providers” from the Volunteer Management System Non-Functional Requirements section
- Removed “Requirements Validation” section and replaced it with “Change Record”
- Added “Employer Identification Number” to the Organization Portal Functional Requirements section
- Removed “Instagram Account Link” from the Organization Portal Functional Requirements section
- Removed “Facebook Account Link” from the Organization Portal Functional Requirements section
- Removed “X Account Link” from the Organization Portal Functional Requirements section
- Added “Add service areas” to the Admin User Functional Requirements section
- Added “Update service areas” to the Admin User Functional Requirements section
- Added “Delete service areas” to the Admin User Functional Requirements section
- Added a link to the `ServiceArea.label` field to the Organizational Portal section.